



SOP: Purchase Order / Goods Receipt / Supplier Invoice Reconciliation

SOP ID: SOP_PO_GR_SI_RECON_V2

Exception Type: Three-Way Match Discrepancy

Step 1: Confirm Exception Context

This SOP applies when three procurement documents are submitted together for reconciliation: a Purchase Order (PO), one or more Goods Receipt Notes (GRN/GR), and one or more Supplier Invoices (SI). The agent must verify that all three documents relate to the same transaction before performing any field checks.

- Confirm all documents reference the same Purchase Order.
- Multiple GR or SI documents may be present; treat quantities as cumulative totals.
- A Supplier Number is mandatory on the PO and GR; it is optional (non-blocking) on the SI.

Field Alias Rule (applies to ALL documents): Document layouts vary by vendor and ERP system. Each field below lists its accepted aliases. When extracting, scan the entire document for ANY of the listed label variants — the field name used internally in the extracted JSON must always be the canonical name shown in bold, regardless of which alias was found in the source document.

Step 2: Fields to Extract per Document

2.1 Purchase Order (PO)

All fields are **mandatory**. Set value to NOT_FOUND if the field cannot be located under any of its aliases.

Canonical Field Name + Accepted Aliases	Mandatory	Notes
supplier_number <i>Also: Vendor No / Vendor Number / Vendor ID / Supplier ID / Supplier No / Creditor No / Party No</i>	Yes	Vendor / supplier identifier on the PO
PO_number <i>Also: PO No / PO Number / Purchase Order No / Purchase Order Number / Order No / Order Number / Purchasing Doc / PO Ref</i>	Yes	Purchase Order reference number
PO_amount <i>Also: PO Value / Order Value / Total Value / Net Value / Total Amount / Order Amount / PO Total</i>	Yes	Total monetary value of the PO
material_quantity <i>Also: Qty / Quantity / Order Qty / Ordered Qty / Total Qty / PO Quantity / Ordered Quantity</i>	Yes	Sum of all line-item quantities on the PO
vendor_name <i>Also: Vendor Name / Supplier Name / Creditor Name / Party Name / Sold By / Payee</i>	Yes	Name of the vendor / supplier
cost_center_project <i>Also: WBS / WBS Code / WBS Element / Cost Centre / Cost Center / Project Code / CC Code / Internal Order</i>	Yes	WBS code or cost centre reference

2.2 Goods Receipt (GR / GRN)

Multiple GR documents may be present (goods_receipt_1, goods_receipt_2, ...). Extract the fields below from each individually using the aliases to locate them.

Canonical Field Name + Accepted Aliases	Mandatory	Notes
supplier_number <i>Also: Vendor No / Vendor Number / Vendor ID / Supplier ID / Supplier No / Creditor No / Party No</i>	Yes	Must match the PO supplier_number
PO_number <i>Also: PO No / PO Number / Purchase Order No / Purchase Order Number / Order No / Order Number / Purchasing Doc / PO Ref / Reference No / Ref No / Doc No / Document Number</i>	Yes	Must match PO and SI PO_number. Note: GR documents often label the PO reference as "Reference No", "Doc No", or "Order No" — check all aliases before marking NOT_FOUND.
gr_quantity <i>Also: Received Qty / Qty Received / GR Qty / Delivered Qty / Quantity Received / Qty / Quantity</i>	Yes	Quantity received on this GR document

■ **Extraction Warning — PO Number on Delivery / GR Documents:** The PO Number may not appear as "PO Number" on GRNs or delivery receipts. It is commonly labelled as *Reference No*, *Ref No*, *Doc No*, *Document Number*, *Order No*, or *Purchasing Doc*. Always search all alias labels across the full document — including headers, footers, and reference fields — before concluding the value is NOT_FOUND.

2.3 Supplier Invoice (SI)

Multiple SI documents may be present (supplier_invoice_1, supplier_invoice_2, ...). Extract from each. supplier_number is optional — do not fail if absent.

Canonical Field Name + Accepted Aliases	Mandatory	Notes
PO_number <i>Also: PO No / PO Number / Purchase Order No / Purchase Order Number / Order No / Order Number / Our Order No / Ref PO / PO Ref / Customer PO / Client PO / Reference No</i>	Yes	Must match PO and GR PO_number
gr_quantity <i>Also: Qty / Quantity / Billed Qty / Invoice Qty / Qty Supplied / Units / No. of Units</i>	Yes	Quantity billed on this invoice
invoice_number <i>Also: Invoice No / Invoice Number / Inv No / Bill No / Bill Number / Tax Invoice No / Document No</i>	Yes	Unique invoice reference
invoice_amount <i>Also: Invoice Amount / Invoice Value / Total Amount / Net Amount / Bill Amount / Amount Due / Payable Amount / Grand Total</i>	Yes	Total monetary value on this invoice
supplier_number <i>Also: Vendor No / Vendor Number / Vendor ID / Supplier ID / Supplier No / Creditor No / Our Ref / Seller ID</i>	No	Extract if present; do not fail if missing

Step 3: Field Comparison Rules

Once all fields are extracted (using their canonical names), perform the three checks below in order. All three must pass for overall reconciliation status to be 'yes'.

Check	Fields Compared	Documents Involved	Pass Condition
1. Supplier Number Match	supplier_number	PO vs GR (SI excluded)	PO supplier_number == GR supplier_number for every GR document. SI supplier_number is N/A for this check.
2. PO Number Match	PO_number	PO vs GR vs SI	PO_number must be identical across the PO, every GR, and every SI. Use aliases to locate the value; always store under canonical name PO_
3. Quantity Match	gr_quantity / material_quantity	GR (total) vs SI (total)	SUM of gr_quantity across all GR docs == SUM of gr_quantity across all SI docs. PO material_quantity is reference only — NOT included in this comparison.

Step 4: Numeric Comparison Rules

- Ignore trailing decimal zeros: 1600, 1600.0, 1600.00 are equal.
- Compare only the significant numeric value, not string representation.
- Strip currency symbols and thousand-separator commas before comparison.

Step 5: Classify Severity

Failing Check	Severity	Reason
supplier_number mismatch (PO vs GR)	High	Wrong vendor may have delivered goods
PO_number mismatch across any document	High	Documents do not belong to the same transaction
Quantity mismatch (GR total vs SI total)	High	Overbilling or underbilling detected
supplier_number missing on SI only	Low / Info	Invoice may not carry vendor ID; not a blocking error
NOT_FOUND for any mandatory field	High	Document is incomplete or wrong file uploaded — check aliases before flagging

Step 6: Generate Reasoning Summary

For each failing check, produce a plain-language explanation covering:

- Which field failed and on which documents.
- The actual values that were compared (including which alias label was matched).
- Why the mismatch matters (financial, operational, or compliance impact).

Example tone:

"PO-12345 shows a quantity of 500 units. Combined GR quantity (found under label "Received Qty") is 480 units; combined SI quantity is 500 units. GR total does not match SI total — possible short delivery or billing discrepancy."

Step 7: Dashboard Output Structure

Populate the output payload with the following fields:

```
{
  "exception_type": "Three-Way Match Discrepancy",
  "severity": "High",
  "recommended_action": "Review mismatched fields and correct source documents.",
  "agent_reasoning_summary": "",
  "status": "Pending",
```

```
"confidence_score": 0.92
}
```

Step 8: Recommended Actions

Check	Pass	Fail — Recommended Action
Supplier Number (PO vs GR)	Record and proceed.	Flag for vendor master review. Verify correct vendor delivered goods.
PO Number (PO / GR / SI)	Record and proceed.	Reject batch. Request correct documents referencing the same PO.
Quantity (GR total vs SI total)	Record and proceed.	Raise discrepancy note. Request credit memo or corrected invoice.